



Connect images with data

Hello again. Earlier we talked about why data visualizations are so important to both analysts and stakeholders. Now we'll discuss the connections you can make between data and images in your visualizations. Visual communication of data is important to those using the data to help make decisions. To better understand the connection between data and images, let's talk about some examples of data visualizations and how they can communicate data effectively. You maybe come across lots of these in your daily life. We'll explore them a little bit more here.

A good place to start is a bar graph. Bar graphs use size contrast to compare two or more values. The horizontal line of a bar graph usually placed at the bottom, is called the x-axis, and bar graphs with vertical bars, the x-axis is used to represent categories, time periods, or other variables. The vertical line of a bar graph usually placed to the left is called the y-axis. The y-axis usually has a scale of values for the variables. In this example, the time of day is compared to someone's level of motivation throughout the whole workday. Bar graphs are a great way to clarify trends.

Here, it's clear this person's motivation is low at the beginning of the day and gets higher and higher by the end of the workday. This type of visualization makes it very easy to identify patterns. Another example is a line graph. Line graphs are a type of visualization that can help your audience understand shifts or changes in your data.

They're usually used to track changes through a period of time, but they can be paired with other factors too. In this line graph, we're using two lines to compare the popularity of cats and dogs over a period of time. With two different line colors, we can immediately tell that dogs are more popular than cats.

We'll talk more about using colors and patterns to make visualizations more accessible to audiences later too. Even as a line moves up and down, there's a general trend upwards and the line for dogs always stays higher than the line for cats. Now let's check out another visualization you'll probably recognize. Say hello to the pie chart. Pie charts show how much each part of something makes up the whole. This pie chart shows us all the activities that make up someone's day. Half of it's spent working, which is shown by the amount of space that the blue section takes up.

From a quick scan, you can easily tell which activities make up a good chunk of the day in this pie chart and which ones take up less time. Earlier, we learned how maps help organize data geographically. The great thing about maps is they can hold a lot of location-based information and they're easy for your audience to interpret. This example shows survey data about people's happiness in Europe. The borderlines are well-defined and the colors added make it even easier to tell the countries apart. Understanding the data represented here, which we'll come back to again later, can happen pretty quickly. So data visualization is an excellent tool for making the connection between an image and the information it represents, but it can sometimes be misleading.

One way visualizations can be manipulated is with scaling and proportions. Think of a pie chart. **Pie charts show proportions and percentages between categories.** Each part of the circle or pie should reflect its percentage to the whole, which is equal to 100 percent. So if you want to visualize your sales analysis to show the percentage of your company sales that come from online transactions, you could use a pie chart. The size of each slice would be the percentage of total sales that it represents. So if your online sales accounted for 60 percent, the slice would be 60 percent of the whole pie.

Now here's a misleading pie chart. It's supposed to show opinions about pizza toppings, but each slice or segment represents more than one option. They all add up to well over 100 percent. There is lots of ingredients listed below the image that are not even included in the visual data. All of the segments are the same size, even though they're supposed to be showing different values. If a visualization looks confusing then it probably is confusing. Let's explore another example where the size of the graphic components comes into play.

This time with a bar chart. In a truncated bar chart like this one, the values on the y-axis don't start at zero. The data points start at 9,100 and at intervals of 100. This makes it seem like the data, let's say, it's for novel clicks per day on different website links, is fairly wide-ranging. In this view, website E seems to clearly receive way more clicks than website D, which receives more clicks than website C and so on. While the graph is clear and the elements are easy to understand, the way the data is presented is misleading. Let's try to fix this by changing the graph's y-axis, so that it starts at zero instead.

Now, the difference between the website clicks per day don't look nearly as drastic. By making the y-axis start at zero, we're changing the visual proportions to be more accurate and more honest. Some platforms always start their y-axis at zero, but other programs like spreadsheets might not fix the y-axis. So it's important to keep this in mind when creating visualizations. By following the conventions of data analysis, you'll be able to avoid misleading visualizations. You always want your visualization to be clear and easy to understand, but never at the expense of communicating ideas that are true to the data. So we've talked about some effective data-driven visualizations like bar graphs, line graphs, and pie charts, and when to use them.

On top of that, we've discussed some things to avoid in your visualizations to keep them from being misleading. Coming up, we'll check out how to make those visualizations reach your target audience. See you then.